

DSA0405
FUNDAMENTALS OF DATA SCIENCE

UNIT 3 EXERCISE PROBLEM

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1. Mean, Median, Mode, and Range

Problem: Find the mean, median, mode and range for the data set: 12, 15, 18, 15, 20, 22, 15, 25.

Concept

These four values are the most basic descriptive statistics used to summarize a data set:

- Mean (arithmetic average) — the sum of all observations divided by the number of observations. It represents the central 'balancing point' of the data.
- Median — the middle value when data is arranged in ascending order. It splits the data into two equal halves and is not affected by extreme values.
- Mode — the value that occurs most frequently in the data set.
- Range — the difference between the maximum and minimum values; it gives a quick, rough idea of spread.

Step 1 — Arrange the Data

Given data (n = 8): 12, 15, 18, 15, 20, 22, 15, 25

Sorted in ascending order: 12, 15, 15, 15, 18, 20, 22, 25

Step 2 — Compute the Mean

$$\text{Mean} = \Sigma X / n$$

$$\Sigma X = 12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

$$\text{Mean} = 142 / 8 = 17.75$$

Step 3 — Compute the Median

Since n = 8 (even), Median = average of the (n/2)th and (n/2 + 1)th terms = 4th and 5th terms

Sorted data position: 1:12 2:15 3:15 4:15 5:18 6:20 7:22 8:25

$$\text{Median} = (15 + 18) / 2 = 33 / 2 = 16.5$$

Step 4 — Identify the Mode

Frequency of each value: 12→1, 15→3, 18→1, 20→1, 22→1, 25→1

The value 15 occurs 3 times — more than any other value

Mode = 15

Step 5 — Compute the Range

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

$$\text{Range} = 25 - 12 = 13$$

Final Answers

Measure	Value
Mean	17.75
Median	16.5
Mode	15
Range	13

Interpretation: The mean (17.75) is slightly higher than the median (16.5), suggesting a mild pull toward higher values (25) — a hint of positive skew even in this small data set. The mode (15) shows that a score of 15 was the most commonly repeated observation.

2. Frequency Distribution

Problem: Construct a frequency distribution for the data: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Concept

A frequency distribution is a tabular summary that shows how many times (frequency) each distinct value occurs in a data set. It is the first step in organizing raw, ungrouped data so that patterns — such as the most common values — become visible at a glance. For discrete data with a small number of distinct values (like this one), we list every unique value once and count its occurrences using tally marks.

Step 1 — List the Raw Data

Data ($n = 12$): 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

Distinct values present: 2, 3, 4, 5, 6

Step 2 — Tally and Count

Value (x)	Tally	Frequency (f)	Relative Frequency (f/n)	Percentage (%)
2		3	$3/12 = 0.25$	25.0%
3		3	$3/12 = 0.25$	25.0%
4		3	$3/12 = 0.25$	25.0%
5		2	$2/12 \approx 0.167$	16.7%
6		1	$1/12 \approx 0.083$	8.3%

Step 3 — Verify the Total

Sum of frequencies = $3 + 3 + 3 + 2 + 1 = 12$

This equals $n = 12$, confirming that every observation has been correctly accounted for.

Step 4 — Cumulative Frequency

Value (x)	Frequency (f)	Cumulative Frequency
2	3	3
3	3	6
4	3	9
5	2	11
6	1	12

Interpretation: The values 2, 3, and 4 are the most common, each occurring 3 times and together making up 75% of the data. The value 6 is the rarest, occurring only once. The distribution is fairly concentrated in the lower range (2–4) and tapers off toward 6, suggesting a mild leftward concentration (right-tailed thinning) in this small sample.

3. Quartiles Q1, Q2, Q3 and Interquartile Range (IQR)

Problem: Find Q1, Q2, Q3 and IQR for the data: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Concept

Quartiles divide an ordered data set into four equal parts, each containing 25% of the observations:

- Q1 (First / Lower Quartile) — the median of the lower half of the data; 25% of values lie below it.
- Q2 (Second Quartile) — this is simply the overall median; 50% of values lie below it.
- Q3 (Third / Upper Quartile) — the median of the upper half of the data; 75% of values lie below it.
- IQR (Interquartile Range) = $Q3 - Q1$ — it measures the spread of the middle 50% of the data and is a robust measure of dispersion, unaffected by extreme values.

Step 1 — Arrange and Confirm Order

Data (n = 10, already sorted): 5, 8, 10, 12, 15, 18, 20, 22, 25, 30

Position: 1:5 2:8 3:10 4:12 5:15 6:18 7:20 8:22 9:25 10:30

Step 2 — Find Q2 (Median)

Since n = 10 (even), Median = average of the 5th and 6th values

$$Q2 = (15 + 18) / 2 = 33 / 2 = 16.5$$

Step 3 — Find Q1 (Lower Quartile)

Lower half (first 5 values, below the median): 5, 8, 10, 12, 15

Q1 = median of the lower half = the middle (3rd) value = 10

Step 4 — Find Q3 (Upper Quartile)

Upper half (last 5 values, above the median): 18, 20, 22, 25, 30

Q3 = median of the upper half = the middle (3rd) value = 22

Step 5 — Compute the IQR

$$IQR = Q3 - Q1$$

$$IQR = 22 - 10 = 12$$

Final Answers

Q1	Q2 (Median)	Q3	IQR
10	16.5	22	12

Interpretation: The middle 50% of the data (the interquartile range) spans from 10 to 22, a width of 12 units. This range is used as the basis for detecting outliers using the $1.5 \times IQR$ rule, applied in the next question.

4. Outlier Detection using the IQR Method

Problem: Detect outliers using the IQR method for the data: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Concept

The IQR (Tukey's) method is a robust technique for flagging outliers — values unusually far from the rest of the data. It defines an acceptable range (the 'fences') from the spread of the middle 50%. Values outside these fences are outliers: Lower Fence = $Q1 - 1.5 \times IQR$, Upper Fence = $Q3 + 1.5 \times IQR$.

Step 1 — Arrange the Data

Data (n = 9, already sorted): 10, 12, 13, 15, 16, 17, 18, 20, 45

Step 2 — Find the Median (Q2)

Since n = 9 (odd), Median = the middle (5th) value
Q2 = 16

Step 3 — Find Q1 and Q3

Lower half (below median): 10, 12, 13, 15 → Q1 = average of 2nd and 3rd values = $(12 + 13)/2 = 12.5$
Upper half (above median): 17, 18, 20, 45 → Q3 = average of 2nd and 3rd values = $(18 + 20)/2 = 19$

Step 4 — Compute the IQR

$IQR = Q3 - Q1 = 19 - 12.5 = 6.5$

Step 5 — Compute the Fences

Lower Fence = $Q1 - 1.5 \times IQR = 12.5 - 1.5(6.5) = 12.5 - 9.75 = 2.75$
Upper Fence = $Q3 + 1.5 \times IQR = 19 + 1.5(6.5) = 19 + 9.75 = 28.75$
Acceptable range: [2.75, 28.75]

Step 6 — Check Each Value Against the Fences

Value	Within [2.75, 28.75]?	Verdict
10	Yes	Normal
12	Yes	Normal
13	Yes	Normal
15	Yes	Normal
16	Yes	Normal
17	Yes	Normal
18	Yes	Normal
20	Yes	Normal
45	No (45 > 28.75)	OUTLIER

Conclusion: The value 45 lies outside the upper fence (28.75), so it is flagged as an OUTLIER; all other values lie comfortably within the fences. A single unusually high value like this can distort the mean and standard deviation, so it should be investigated — it could be a genuine extreme case or a data-entry/measurement error.

5. Skewness (Asymmetry) of the Distribution

Problem: Determine the skewness of the data: 5, 6, 7, 7, 8, 8, 9, 20.

Concept

Skewness measures the asymmetry of a distribution around its mean:

- Zero skewness → the distribution is symmetric (mean = median).
- Positive (right) skew → the tail extends toward higher values; Mean > Median.
- Negative (left) skew → the tail extends toward lower values; Mean < Median.

A common measure is Pearson's Coefficient of Skewness: $\text{Skewness} = 3(\text{Mean} - \text{Median}) / \text{Standard Deviation}$.

Step 1 — Compute the Mean

$$\Sigma X = 5+6+7+7+8+8+9+20 = 70, \quad n = 8$$
$$\text{Mean} = 70 / 8 = 8.75$$

Step 2 — Compute the Median

Sorted data (already sorted): 5, 6, 7, 7, 8, 8, 9, 20
 $n = 8$ (even) → Median = average of 4th and 5th values = $(7 + 8)/2 = 7.5$

Step 3 — Identify the Mode

7 occurs twice, 8 occurs twice — both are the most frequent values
Mode = 7 and 8 (bimodal)

Step 4 — Compute the Standard Deviation

Deviation table (using population formula, mean = 8.75):

X	X – Mean	(X – Mean) ²
5	–3.75	14.0625
6	–2.75	7.5625
7	–1.75	3.0625
7	–1.75	3.0625
8	–0.75	0.5625
8	–0.75	0.5625
9	0.25	0.0625
20	11.25	126.5625
$\Sigma(X - \text{Mean})^2 = 14.0625 + 7.5625 + 3.0625 + 3.0625 + 0.5625 + 0.5625 + 0.0625 + 126.5625 = 155.5$		
Population Variance = $\Sigma(X - \text{Mean})^2 / n = 155.5 / 8 = 19.4375$		
Population Standard Deviation = $\sqrt{19.4375} \approx 4.41$		

Step 5 — Compute Pearson's Coefficient of Skewness

$$\text{Skewness} = 3(\text{Mean} - \text{Median}) / \text{SD}$$
$$\text{Skewness} = 3(8.75 - 7.5) / 4.41 = 3(1.25) / 4.41 = 3.75 / 4.41 \approx 0.85$$

Conclusion: Mean (8.75) > Median (7.5), and the skewness coefficient (≈ 0.85) is positive, so the distribution is POSITIVELY (right) SKEWED. The single large value 20 pulls the mean well above the median, stretching the right tail — a classic example of how one extreme observation can visibly skew an otherwise tightly clustered data set (5–9).

6. Continuous Uniform Distribution — $P(20 < X < 35)$

Problem: A uniformly distributed variable ranges from 10 to 50. Find $P(20 < X < 35)$.

Concept

A continuous uniform distribution describes a variable where every value in an interval $[a, b]$ is equally likely to occur. Its probability density function (PDF) is a flat, constant line over the interval:

$$f(x) = 1 / (b - a), \text{ for } a \leq x \leq b$$
$$f(x) = 0, \text{ otherwise}$$

Because the density is constant, the probability of X falling in any sub-interval $[c, d]$ within $[a, b]$ is simply proportional to the length of that sub-interval:

$$P(c < X < d) = (d - c) / (b - a)$$

Step 1 — Identify the Parameters

$$a = 10 \text{ (minimum), } b = 50 \text{ (maximum)}$$
$$\text{Total width of the distribution} = b - a = 50 - 10 = 40$$

Step 2 — Set Up the Required Probability

$$\text{We need } P(20 < X < 35), \text{ i.e., } c = 20 \text{ and } d = 35$$
$$\text{Width of the required interval} = d - c = 35 - 20 = 15$$

Step 3 — Apply the Uniform Probability Formula

$$P(20 < X < 35) = (35 - 20) / (50 - 10) = 15 / 40$$
$$P(20 < X < 35) = 0.375$$

Additional Useful Quantities for this Distribution

Quantity	Formula	Value
Mean, μ	$(a + b) / 2$	$(10+50)/2 = 30$
Variance, σ^2	$(b - a)^2 / 12$	$(40)^2 / 12 \approx 133.33$
Standard Deviation, σ	$\sqrt{\text{Variance}}$	≈ 11.55
$P(X < 20)$	$(20 - 10) / 40$	0.25
$P(X > 35)$	$(50 - 35) / 40$	0.375

Final Answer

$$P(20 < X < 35) = 0.375 \text{ (equivalently, 37.5\%)}$$

Interpretation: Because probability is proportional to interval length in a uniform distribution, the 15-unit-wide interval $[20, 35]$ captures exactly $15/40 = 37.5\%$ of the total probability spread evenly across $[10, 50]$.

7. Z-Score for $X = 130$ in a Normal Distribution

Problem: A normal distribution has mean 100 and standard deviation 15. Find the Z-score for $X = 130$.

Concept

The Z-score (standard score) tells us how many standard deviations an observation X lies from the mean μ of a normal distribution. Converting X into a Z-score allows us to use the standard normal distribution table (which has $\mu = 0$, $\sigma = 1$) to find probabilities for any normal distribution.

$$Z = (X - \mu) / \sigma$$

Step 1 — List the Given Values

μ (mean) = 100
 σ (standard deviation) = 15
 X (observation) = 130

Step 2 — Apply the Z-Score Formula

$$\begin{aligned} Z &= (X - \mu) / \sigma \\ Z &= (130 - 100) / 15 \\ Z &= 30 / 15 \\ Z &= 2 \end{aligned}$$

Step 3 — Interpret the Result

A Z-score of +2 means that $X = 130$ lies exactly 2 standard deviations ABOVE the mean.
By the Empirical Rule (68–95–99.7 rule), about 95% of all values in a normal distribution lie within ± 2 standard deviations of the mean.
Therefore $X = 130$ lies at the edge of the 'typical' 95% range — it is a relatively high, though not extreme, value.

Final Answer

$$Z = 2$$

Step 4 — The Empirical (68–95–99.7) Rule

The empirical rule provides a quick sanity check for Z-scores in a normal distribution, describing what proportion of data falls within each band of standard deviations from the mean:

Interval	Z-range	Approx. % of Data
$\mu \pm 1\sigma$	–1 to +1	$\approx 68\%$
$\mu \pm 2\sigma$	–2 to +2	$\approx 95\%$
$\mu \pm 3\sigma$	–3 to +3	$\approx 99.7\%$

Note: Since $Z = 2$ for $X = 130$, this value sits exactly at the boundary of the $\mu \pm 2\sigma$ band, which covers about 95% of the distribution. This Z-score will be used directly in the next question to compute the cumulative probability $P(X < 130)$ using the standard normal table.

8. Cumulative Probability $P(X < 130)$

Problem: For the same normal distribution ($\mu = 100$, $\sigma = 15$), find $P(X < 130)$.

Concept

To find the probability that a normally distributed variable X is less than some value, we first convert X to its Z -score (as in Question 7), then look up the cumulative probability $P(Z < z)$ in the standard normal distribution table. This table gives the area under the standard normal curve to the left of a given Z value.

Step 1 — Recall the Z-Score

From Question 7: $Z = (X - \mu) / \sigma = (130 - 100) / 15 = 2$

Step 2 — Look Up the Standard Normal Table

$P(X < 130) = P(Z < 2)$

From the standard normal (Z) table, the cumulative area to the left of $Z = 2.00$ is 0.9772

Step 3 — State the Result

$P(X < 130) = 0.9772$ (i.e., 97.72%)

Related Probabilities (for context)

Event	Calculation	Probability
$P(X < 130)$	$P(Z < 2)$	0.9772
$P(X > 130)$	$1 - P(Z < 2)$	0.0228
$P(70 < X < 130)$	$P(-2 < Z < 2)$	≈ 0.9545

Interpretation: There is a 97.72% chance that a randomly selected value from this distribution will be less than 130. Equivalently, only about 2.28% of values exceed 130, confirming that $X = 130$ is near the upper end of the typical range of the distribution.

How the Standard Normal Table Works

The standard normal (Z) table lists the cumulative area under the standard normal curve to the left of a given Z -value. To read it: find the row for the Z -value rounded to one decimal (2.0), then the column for the second decimal (0.00), giving the intersection value 0.9772. Because the total area under any probability curve equals 1, the area to the right of Z is simply 1 minus the table value.

Summary

Given: $\mu = 100$, $\sigma = 15$, $X = 130 \rightarrow Z = 2$
 $P(X < 130) = P(Z < 2) = 0.9772$ (97.72%)
 $P(X > 130) = 1 - 0.9772 = 0.0228$ (2.28%)

9. Effect of an Outlier on Mean vs Median

Problem: Given a data set with one extreme value, compare the mean and median before and after outlier treatment.

Concept

The mean uses every value in its calculation, so a single very large or very small value can pull it substantially in that direction. The median, on the other hand, depends only on the position of values in the sorted order — not their magnitude — so it is far more resistant ('robust') to outliers. This question illustrates that difference using a concrete data set.

Data Set Used

10, 12, 13, 15, 16, 17, 18, 20, 45 (n = 9) — the value 45 is a clear extreme value, as established in Question 4.

Step 1 — Statistics BEFORE Removing the Outlier

Data: 10, 12, 13, 15, 16, 17, 18, 20, 45 (n = 9)
Mean = $\Sigma X / n = (10+12+13+15+16+17+18+20+45) / 9 = 166 / 9 \approx 18.44$
Median = middle (5th) value of the sorted data = 16

Step 2 — Statistics AFTER Removing the Outlier

Remaining data: 10, 12, 13, 15, 16, 17, 18, 20 (n = 8)
Mean = $\Sigma X / n = (10+12+13+15+16+17+18+20) / 8 = 121 / 8 = 15.125$
Median = average of 4th and 5th values = $(15 + 16) / 2 = 15.5$

Step 3 — Side-by-Side Comparison

Statistic	Before (with 45)	After (without 45)	Absolute Change	% Change
Mean	18.44	15.125	3.32	$\approx 18.0\%$ decrease
Median	16	15.5	0.5	$\approx 3.1\%$ decrease

Step 4 — Why the Difference Occurs

- The mean is calculated from the sum of all values, so the single large value 45 contributes disproportionately to the total, inflating the mean well above where most of the data actually lies.
- The median only looks at the middle position in the sorted list. Removing 45 shifts which values fall in the middle, but only by a small amount, since 45 was simply the topmost value regardless of its actual size.

Conclusion

The mean changed by about 3.32 points ($\approx 18\%$), a substantial shift.
The median changed by only 0.5 points ($\approx 3\%$), a minor shift.
This confirms that the MEDIAN is a more ROBUST (outlier-resistant) measure of central tendency than the MEAN.

Practical takeaway: When a data set is suspected to contain extreme values or errors (e.g., income data, response times, sensor glitches), the median often gives a more representative picture of the 'typical' value than the mean.

10. Complete Exploratory Data Analysis (EDA) — Student Marks

Problem: Perform a complete EDA on a small student-mark data set and report central tendency, dispersion, distribution, outliers and skewness.

Concept

Exploratory Data Analysis (EDA) is the process of summarizing a data set's main characteristics, often using statistical measures, before applying more formal modelling. A complete EDA typically covers: central tendency (where the data is centred), dispersion (how spread out it is), the shape of the distribution (skewness), and the presence of outliers.

Data Set Used

Marks out of 100 for 10 students (sorted): 45, 50, 55, 60, 62, 65, 70, 72, 75, 95 (n = 10)

a) Central Tendency

Mean = $\Sigma X / n = (45+50+55+60+62+65+70+72+75+95) / 10 = 649 / 10 = 64.9$

Median = average of 5th and 6th values = $(62 + 65) / 2 = 63.5$

Mode = None — all ten values are distinct

b) Dispersion

Deviation table for variance and standard deviation (Mean = 64.9):

X	X – Mean	(X – Mean) ²
45	–19.9	396.01
50	–14.9	222.01
55	–9.9	98.01
60	–4.9	24.01
62	–2.9	8.41
65	0.1	0.01
70	5.1	26.01
72	7.1	50.41
75	10.1	102.01
95	30.1	906.01

$\Sigma(X - \text{Mean})^2 = 396.01+222.01+98.01+24.01+8.41+0.01+26.01+50.41+102.01+906.01 = 1832.9$

Population Variance = $1832.9 / 10 = 183.29$

Population Standard Deviation = $\sqrt{183.29} \approx 13.54$

Range = Max – Min = $95 - 45 = 50$

Q1 (lower half: 45,50,55,60,62) = 55 | Q3 (upper half: 65,70,72,75,95) = 72

IQR = Q3 – Q1 = $72 - 55 = 17$

c) Five-Number Summary

Minimum	Q1	Median (Q2)	Q3	Maximum
45	55	63.5	72	95

d) Distribution Shape / Skewness

Mean (64.9) is slightly greater than Median (63.5) → Mean > Median

Pearson's Coefficient of Skewness = $3(\text{Mean} - \text{Median}) / \text{SD} = 3(64.9 - 63.5) / 13.54 = 3(1.4)/13.54 \approx 0.31$

A small positive coefficient indicates a MILD POSITIVE (right) skew — a few higher marks (notably 95) stretch the right tail of the distribution.

e) Outlier Detection (IQR Method)

Lower Fence = $Q1 - 1.5 \times IQR = 55 - 1.5(17) = 55 - 25.5 = 29.5$
Upper Fence = $Q3 + 1.5 \times IQR = 72 + 1.5(17) = 72 + 25.5 = 97.5$
Acceptable range: [29.5, 97.5]. Every value in the data set (45 to 95) falls within this range.
Conclusion: No statistical outliers are present, although 95 is the highest mark and closest to the upper fence.

f) Overall EDA Summary Table

Aspect	Result
Mean	64.9
Median	63.5
Mode	None
Range	50
IQR	17
Standard Deviation	≈ 13.54
Skewness	Mild positive (right) skew (≈ 0.31)
Outliers (IQR method)	None detected

Overall Conclusion

The student marks are centred around the low-to-mid 60s (mean 64.9, median 63.5), with a moderate spread ($SD \approx 13.5$, $IQR = 17$). The distribution shows a mild rightward skew driven by a few stronger performers (up to 95), but this high mark does not qualify as a statistical outlier. Overall, the class performance is fairly typical and consistent, without any data points requiring special investigation.